

8_popquiz electrostatics

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:08pm

Quiz Instructions



Question 1 20 pts



I touched the top of an electroscope with a balloon after it was rubbed against fur. The leaves are made of thin aluminum foil and are mobile.

What was supposed happened ? hint: which object like electrons the most. fur or balloon? see slides.



electrons are transferred to the top and stay there. The leaves don't move



electrons are transferred to the electroscope and spread every where. The leaves separate.



electrons are transferred to the balloon and the electroscope is charged positively



both the electroscope and the balloon stay neutral



Question 2 20 pts

If you rub a glass rod with silk (see lecture or slide)


☐

It depends on the color of tie.

☐

nothing happened

☐

the glass rod becomes negatively charged

☐

silk becomes negatively charged



Question 3 20 pts

A metallic object holds a charge of -3.8×10^{-6} C. What total number of electrons does this represent? ($e = 1.6 \times 10^{-19}$ C is the magnitude of the electronic charge.)

☐

4.2×10^{14}

☐

6.1×10^{13}

☐

2.4×10^{13}

☐

1.6×10^{14}



Question 4 20 pts

You are a conductor. If you run your feet against the carpet, the electrons will tend to

<https://phet.colorado.edu/en/simulations/john-travoltage>

☐

stay attached to the carpet and will not transfer to your body

☐

will diffuse in the air after being pulled out from the carpet

☐

transfer to your body and spread on the outside of the body

☐

transfer to your body and build up inside your body



Question 5 20 pts

https://phet.colorado.edu/sims/html/balloons-and-static-electricity/latest/balloons-and-static-electricity_en.html  (https://phet.colorado.edu/sims/html/balloons-and-static-electricity/latest/balloons-and-static-electricity_en.html)

After rubbing the ballong against the sweater. What is true

☐

the sweater becomes positively charged

☐

the sweaters stays neutral

☐

electrons spread out on the balloon

☐

the sweater becomes negaively charged

Not saved

Submit Quiz